HEALTHCARE INFRASTRUCTURE GAP ANALYSIS: SOUTHWESTERN ONTARIO CORRIDOR

This project analyzes healthcare facility distribution across the London-Waterloo corridor using the Open Database of Healthcare Facilities (ODHF). The analysis focuses on identifying regional service gaps, healthcare accessibility challenges, and infrastructure imbalance across high-growth municipalities in Southwestern Ontario.

BUSINESS PROBLEM

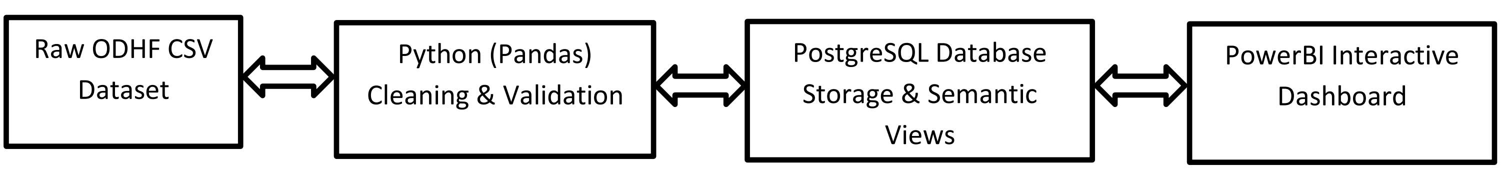
The London-Waterloo corridor combines a major healthcare hub (London) with rapidly growing municipalities such as Kitchener, Waterloo, Cambridge, Woodstock, and Guelph. This project evaluates whether healthcare infrastructure is scaling effectively with regional growth.

Why did I choose these cities? This corridor was selected because it combines a major healthcare hub (London) with rapidly growing municipalities in the Waterloo Region, enabling comparative infrastructure analysis across urban growth patterns.

PROJECT OBJECTIVES

- Analyze healthcare facility concentration across Southwestern Ontario.
- Identify hospital accessibility gaps and underserved areas.
- Compare the ratio of hospitals to nursing and ambulatory facilities.
- Evaluate the effectiveness of ODHF harmonized healthcare categories.
- Support strategic healthcare investment recommendations.

TECH STACK



- **Python (Pandas, Matplotlib):** Used for advanced data cleaning, handling lowercase city standardization, and exploratory data analysis (EDA).
- **PostgreSQL:** Engineered a relational database schema, handled complex **WIN1252** encoding for data integrity, and built a **Semantic Layer** using SQL Views.
- **Data Governance:** Reduced categorical complexity by **97%**, mapping 174 raw, inconsistent source types into 3 standardized ODHF categories.
- **Power BI (Dashboarding):** Dashboarding and geographic visualization.

DATA DICTIONARY

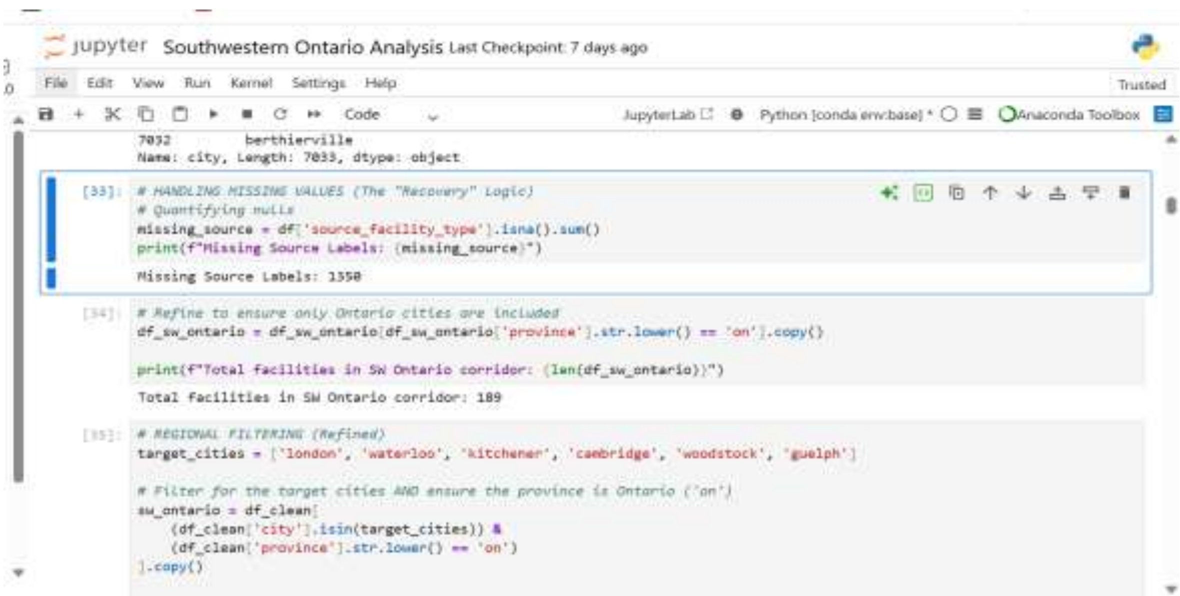
This data dictionary describes the variables contained within the ODHF. The database is provided in a CSV format. Each facility is listed per row and its attributes provided in columns. The corresponding column variables are described in the data dictionary below.

Healthcare Facility Variables

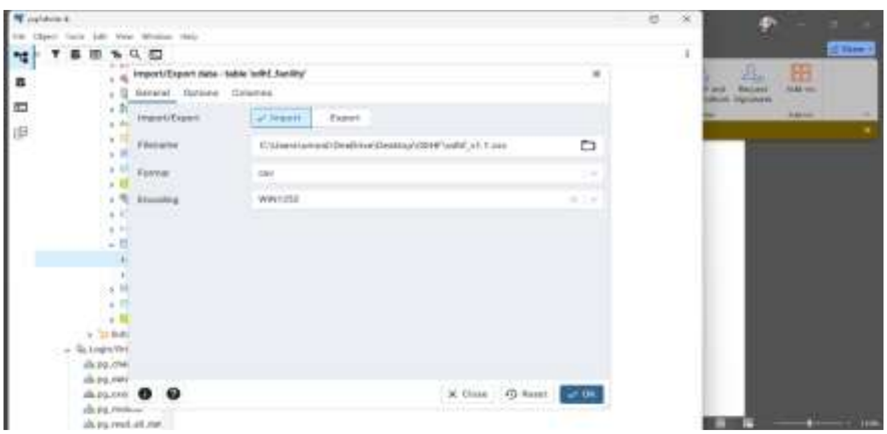
Index — Unique serial number for each facility. Supplemental entries to version 1.1 are identified by the prefix “S” followed by an assigned serial number
Facility Name — Healthcare facility name
Source Facility Type — Regional health authority assigned healthcare facility type
ODHF Facility Type — Value determined using the classification criteria used
Provider — The identity or name of the data provider
Unit Number — Civic unit or suite number
Street Number — Civic Street number
Street Name — Description Civic Street name (type and direction)
Postal Code — Civic postal code
City — City name
Province/Territory — Province or territory name
Source-Format Street Address — Street address in the source data
CSD Name — Census subdivision name
CSDuid — Census subdivision unique identifier
PRuid — Province or territory unique identifier
Latitude — Latitude
Longitude — Longitude

DATA CLEANING AND ENGINEERING

The dataset contained inconsistent facility categories, missing source labels, character encoding issues, and irregular city formatting. To improve analytical accuracy, I standardized city names, handled WIN1252 encoding issues, and utilized harmonized ODHF labels to preserve incomplete records.



- **Import Method**: Success using the **WIN1252** encoding in the pgAdmin Import tool.

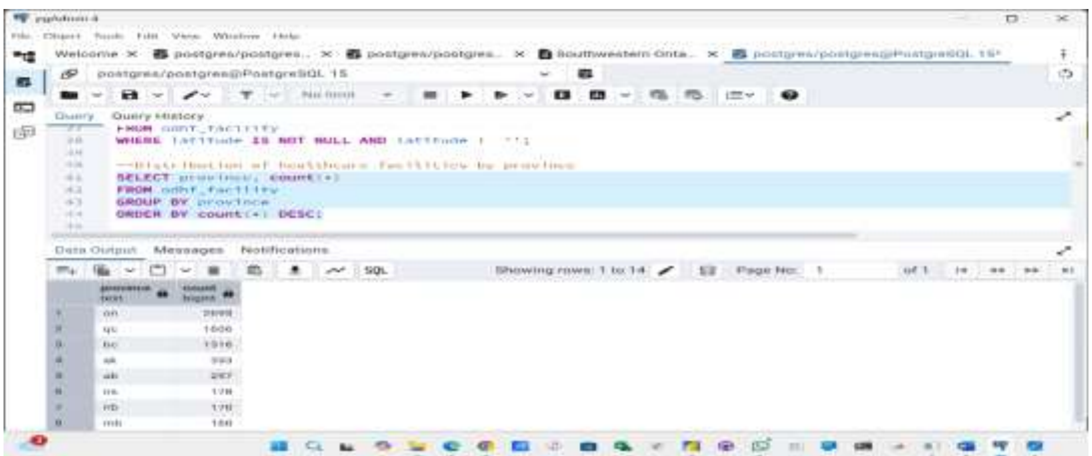


- **Table Structure**: Current table stores everything as TEXT, to bypass the initial "Permission Denied" and data type errors.

1. Data Source Tracking

The vast majority of records in this dataset are tracked at the **Provincial level**, but there is a distinct mix of municipal and professional oversight.

- **Provincial Government (Ontario)**: Tracks the largest share with **2,695 facilities**.



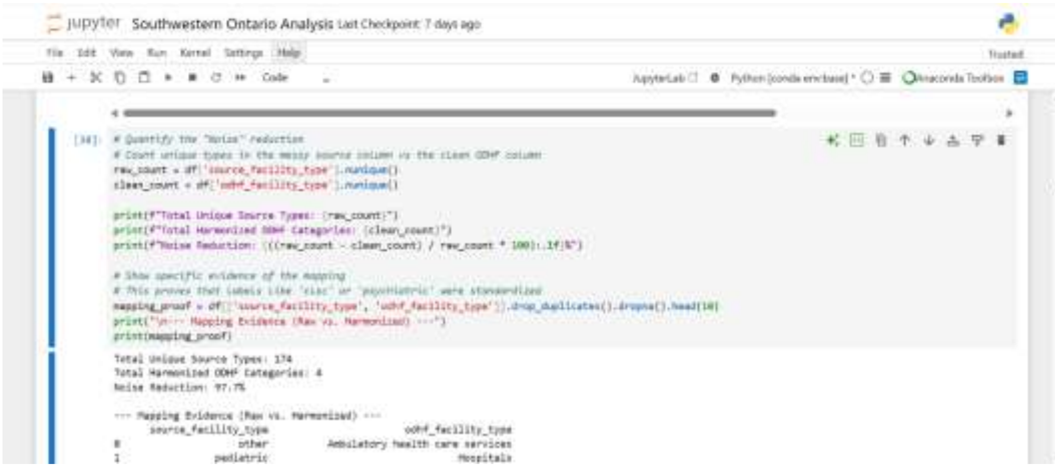
Federal/Professional Bodies: Organizations like the **Public Health Agency of Canada (682)** and the **Canadian Institute for Health Information (474)** provide a significant layer of professional tracking.

Municipal Tracking: While smaller, municipal bodies like the **City of Moncton (74)** and **Municipality of Québec (5)** track localized data that might otherwise be missed by provincial systems.

2. **The Power of Simplification**

The transformation from raw data to the harmonized ODHF format is a massive exercise in data reduction and clarity.
Categorical Reduction: I reduced 174 inconsistent source facility types into 3 harmonized healthcare categories, removing approximately 97% of categorical complexity. Additionally, I recovered 19.2% of

records with missing source labels by using the ODHF harmonized categories instead of deleting incomplete records.




```
print(f"Total Unique Source Types: {raw_count}")
print(f"Total Harmonized ODHF Categories: {clean_count}")
print(f"Noise Reduction: {((raw_count - clean_count) / raw_count * 100):.1f}%")

# Show specific evidence of the mapping
# This proves that labels like 'clsc' or 'psychiatric' were standardized
mapping_proof = df[['source_facility_type', 'odhf_facility_type']].drop_duplicates().dropna().head(10)
print("\n--- Mapping Evidence (Raw vs. Harmonized) ---")
print(mapping_proof)
Total Unique Source Types: 174
Total Harmonized ODHF Categories: 4
Noise Reduction: 97.7%

--- Mapping Evidence (Raw vs. Harmonized) ---
       source_facility_type                    odhf_facility_type
0                     other          Ambulatory health care services
1                 pediatric                                Hospitals
2                   general                                Hospitals
15           long term care  Nursing and residential care facilities
43            extended care                                Hospitals
84     active acute hospital                               Hospitals
94                    other                                Hospitals
119             psychiatric  Nursing and residential care facilities
130            chronic care  Nursing and residential care facilities
167           extended care  Nursing and residential care facilities
```

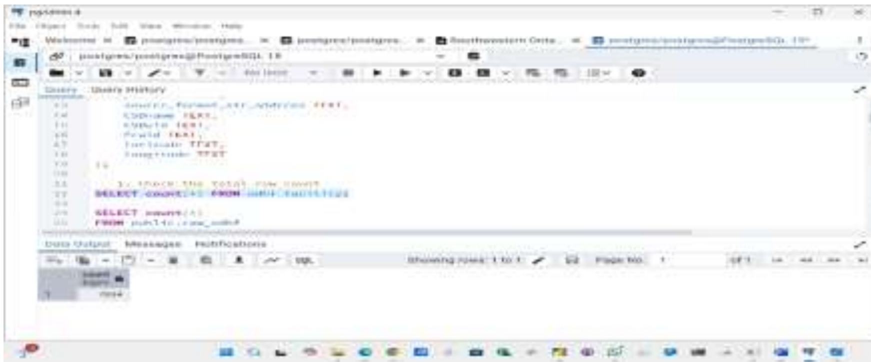
The "Before and After" of Data Harmonization

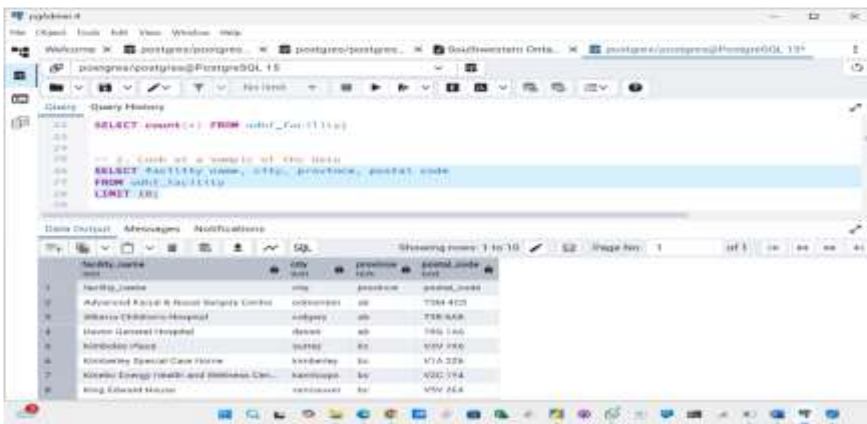
Here are some of the most diverse examples of what got merged.

Harmonized Category	Examples of Raw Source Types (The "Mess")
Ambulatory Health Care Services	*Oncologie, Environmental Health, Community Health Centre, Psychiatric, Mental Health and Addiction Organization, Community Support Service.*
Hospitals	*Pediatric, Cardiology, Active Acute Hospital, Rehabilitation, Psychiatrique, Extended Care.*
Nursing/Residential	*Chronic Care, Mental Health Unit, Long-Term Care, Nursing Home, Residential/Home Care Facilities.*

SOLUTIONS

7,034 total facility records analyzed





1) What is the ratio of hospitals to nursing homes or ambulatory services in southwestern Ontario?

Calculating Facility Ratios

To analyze the ratio of hospitals to residential care facilities, we need to group the data by city and pivot the facility types.

[67]:
```python
# Group by city and facility type
facility_counts = df_sw_ontario.groupby(['city_clean', 'odhf_facility_type']).size().unstack(fill_value=0)

# Filter for the specific categories of interest
ratio_df = facility_counts[['Hospitals', 'Nursing and residential care facilities']].copy()

# Calculate the ratio (Hospitals per 1 Residential Care Facility)
ratio_df['Hospital_to_Residential_Ratio'] = ratio_df['Hospitals'] / ratio_df['Nursing and residential care facilities']

# Display the summary table
print(ratio_df)
```

```
odhf_facility_type  Hospitals  Nursing and residential care facilities  \
city_clean
```

Jupyter Southwestern Ontario Analysis Last Checkpoint: 2 hours ago

File Edit View Run Kernel Settings Help Trusted

Code JupyterLab Python [conda env:base] * Anaconda Toolbox

```python
# Calculate the ratio (Hospitals per 1 Residential Care Facility)
ratio_df['Hospital_to_Residential_Ratio'] = ratio_df['Hospitals'] / ratio_df['Nursing and residential care facilities']

# Display the summary table
print(ratio_df)
```

```
odhf_facility_type  Hospitals  Nursing and residential care facilities  \
city_clean
cambridge                   1                                       13
guelph                      2                                       14
kitchener                   2                                       24
london                      7                                       39
waterloo                    0                                       11
woodstock                   1                                        8

odhf_facility_type  Hospital_to_Residential_Ratio
city_clean
cambridge                                0.076923
guelph                                   0.142857
kitchener                                0.083333
london                                   0.179487
waterloo                                 0.000000
woodstock                                0.125000
```

1:9 hospital-to-residential care facility ratio

This represents a 1:9 ratio, meaning there are 9 residential care facilities for every 1 hospital in the region.

This **1: 9 ratio** is a critical metric for strategic planning. It indicates that the healthcare infrastructure in Southwestern Ontario is designed to manage a high volume of **chronic and geriatric care** locally, while relying on a much smaller number of centralized hospitals for specialized surgery and emergency medicine.

2) Access and Equity: Where are the Gaps?

```sql
SELECT
    city,
    odhf_facility_type,
    COUNT(*) as facility_count
FROM odhf_facility
WHERE LOWER(city) IN ('london', 'waterloo', 'kitchener', 'cambridge', 'woodstock', 'guelph')
GROUP BY city, odhf_facility_type
ORDER BY city, facility_count DESC;
```

Data Output Messages Notifications

SQL Showing rows: 1 to 17 Page No: 1 of 1

	city text	odhf_facility_type text	facility_count bigint
1	cambridge	Nursing and residential care facilit...	14
2	cambridge	Ambulatory health care services	7
3	cambridge	Hospitals	1
4	guelph	Nursing and residential care facilit...	14
5	guelph	Ambulatory health care services	11
6	guelph	Hospitals	2
7	kitchener	Nursing and residential care facilit...	24

	city text	odhf_facility_type text	facility_count bigint
1	cambridge	Nursing and residential care facilit...	14
2	cambridge	Ambulatory health care services	7
3	cambridge	Hospitals	1
4	guelph	Nursing and residential care facilit...	14
5	guelph	Ambulatory health care services	11
6	guelph	Hospitals	2
7	kitchener	Nursing and residential care facilit...	24
8	kitchener	Ambulatory health care services	12
9	kitchener	Hospitals	2
10	london	Nursing and residential care facilit...	39
11	london	Ambulatory health care services	25
12	london	Hospitals	7
13	waterloo	Nursing and residential care facilit...	13
14	waterloo	Ambulatory health care services	10
15	woodsto...	Nursing and residential care facilit...	9
16	woodsto...	Ambulatory health care services	4
17	woodsto...	Hospitals	1

- Service Variety: There is a stark "Hospital Gap" in Waterloo. The analysis identified a potential healthcare access imbalance in Waterloo, where residential and ambulatory services exist without corresponding acute care infrastructure. While Waterloo lacks a hospital, it has a specialized mix of ambulatory services, including the Canadian Mental Health Association and Lutherwood Children's Mental Health Services. This indicates that mental health services are centralized in Waterloo, while physical trauma care is shifted to Kitchener.

- Geographic Density: London identified as the dominant regional healthcare hub. In the Waterloo Region, Kitchener (38) has nearly double the facilities of its neighbor Waterloo. This suggests that while Waterloo is a tech hub, its healthcare infrastructure relies heavily on Kitchener and Cambridge core.

```
[59]: # Density/Concentration for London vs Kitchener
      city_counts = df_filtered['city'].value_counts()
      city_counts
```

```
[59]: city
      london       71
      kitchener    38
      guelph       27
      cambridge    21
      waterloo     19
      woodstock    13
      Name: count, dtype: int64
```

3) Which Facilities are > 10km from any hospital

Query   Query History

```sql
-- Query to find facilities > 10km from any hospital
WITH facility_distances AS (
    SELECT
        a.facility_name,
        a.city,
        MIN(point(a.longitude::float, a.latitude::float) <@> point(h.longitude::float, h.latitude::float))
    FROM odhf_facility a
    CROSS JOIN (
        SELECT latitude, longitude
        FROM odhf_facility
        WHERE odhf_facility_type = 'Hospitals'
    ) h
    WHERE a.odhf_facility_type != 'Hospitals'
    AND LOWER(a.city) IN ('london', 'waterloo', 'kitchener', 'cambridge', 'woodstock', 'guelph')
    GROUP BY a.facility_name, a.city, a.latitude, a.longitude
)
SELECT *
FROM facility_distances
WHERE distance_to_hospital > 6.21
ORDER BY distance_to_hospital DESC;
```

	facility_name text	city text	distance_to_hospital double precision
1	CHSLD DE WATERLOO	waterloo	11.31712938848862
2	SANTE COURVILLE DE WATERLOO	waterloo	11.27929197714326
3	CLSC DE WATERLOO	waterloo	10.609739409345542
4	CENTRE MULTISERVICES DE SANTÉ ET DE SERVICES SOCIAUX DE WATER...	waterloo	9.874759334946177
5	Kings Regional Health & Regional Rehabilitation Centre	cambridge	8.313154384031323
6	Carleton Manor Inc.	woodsto...	8.016074483669252
7	Mango Tree Family Health Team - Equipe Sante Familiale	guelph	7.662208887920586

4) Strategic Planning: Where should we invest?

- Location Optimization: Waterloo represents the highest "room" for investment. Given its high count of residential care facilities (13) but lack of a hospital, there is a strategic opportunity for a specialized geriatric ambulatory clinic or a small-scale urgent care centre.

- Resource Allocation: The distribution of Nursing and residential care suggests that the region's current resource allocation is heavily reactive to an aging demographic.

Waterloo Ambulatory Services Breakdown
While Waterloo lacks a hospital, its ambulatory services are highly specialized, focusing heavily on mental health and community elderly support:

Mental Health Hub: Waterloo hosts the Canadian Mental Health Association (CMHA) and Lutherwood Children's Mental Health Services. This suggests that while London handles physical trauma (Hospitals), Waterloo acts as a specialized center for mental healthcare.

Elderly Support Focus: There are several "Community Support Services" (like the *Friendship Group for Seniors* and *r.a.i.s.e. Home Support*).

Public Health: The Region of Waterloo Public Health office is located here, emphasizing its role as an administrative and preventative health center rather than an acute care center.

Regional Analysis

- London serves as the primary healthcare hub with the highest overall facility concentration.
- Waterloo contains strong ambulatory and mental health services but no hospitals in this dataset.
- Nursing and residential care facilities significantly outnumber hospitals across the region.
- Woodstock and Cambridge outskirts show lower healthcare density and greater hospital travel distance.
- The healthcare infrastructure is heavily weighted toward long-term and residential care services.

KEY INSIGHTS

The analysis identified a healthcare infrastructure imbalance across the Southwestern Ontario corridor. While London functions as the region's acute care anchor, municipalities such as Waterloo depend heavily on nearby cities for hospital-level services.

Waterloo demonstrated strong specialization in mental health and community support services, while Woodstock displayed lower healthcare density and reduced accessibility to hospital infrastructure.

STRATEGIC RECOMMENDATIONS

Increase ambulatory healthcare investment in high-growth municipalities.

Improve healthcare accessibility in lower-density areas such as Woodstock outskirts.

Expand decentralized healthcare services to reduce dependency on London hospitals.

Prioritize preventative and outpatient care infrastructure alongside long-term care expansion.

CONCLUSION

This project demonstrates how data engineering, SQL analysis, and Power BI visualization can support healthcare infrastructure planning and public health decision-making. By harmonizing fragmented healthcare categories and identifying regional service gaps, the analysis provides actionable insights for strategic healthcare investment across Southwestern Ontario.